



Continuum Mechanics

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Chapter 02

Tensor Calculus

Part 3



The Localization Theorem



Result 2.19 (Localization Theorem)

Consider an open set $B \subseteq \mathbb{E}^3$ and a continuous function

$$\phi : B \longrightarrow \mathbb{R}.$$

Let Ω denote an arbitrary open subset of B . If

$$\int_{\Omega} \phi(\mathbf{x}) \, dV = 0 \quad \text{for every open } \Omega \subseteq B, \quad (2.1)$$

then

$$\phi(\mathbf{x}) = 0 \quad \text{for every } \mathbf{x} \in B. \quad (2.2)$$

The word *every* is essential. Cancellation may make an integral zero over one particular region even when ϕ is not identically zero.



Proof.

Assume that (2.1) is true, but (2.2) is false. Then there is some point $\mathbf{y} \in B$ for which

$$\phi(\mathbf{y}) \neq 0.$$

Suppose that

$$\phi(\mathbf{y}) = 2\delta, \quad \delta > 0.$$

The case $\phi(\mathbf{y}) < 0$ can be handled similarly.

Since ϕ is continuous, there is an open neighborhood Ω of $\mathbf{y}$ such that

$$\mathbf{y} \in \Omega \subseteq B$$

and

$$\phi(\mathbf{x}) > \delta, \quad \mathbf{x} \in \Omega.$$



Proof (Cont.)

For this open set Ω ,

$$\int_{\Omega} \phi(\mathbf{x}) \, dV > \delta \, \text{Vol}(\Omega) > 0.$$

This contradicts (2.1), which asserts that

$$\int_{\Omega} \phi(\mathbf{x}) \, dV = 0$$

for every open $\Omega \subseteq B$. Hence (2.2) is true:

$$\phi(\mathbf{x}) = 0, \quad \mathbf{x} \in B.$$



The proof converts information about *every region* into information at *every point*.

Vector and Tensor Localization



The scalar theorem applies component by component. For a continuous vector field $\mathbf{g} : B \rightarrow \mathcal{V}$,

$$\int_{\Omega} \mathbf{g} \, dV = \mathbf{0} \quad \text{for every open } \Omega \subseteq B \quad \implies \quad \mathbf{g} = \mathbf{0} \text{ in } B.$$

Indeed, for every frame vector $\mathbf{e}_i$,

$$0 = \mathbf{e}_i \cdot \int_{\Omega} \mathbf{g} \, dV = \int_{\Omega} g_i \, dV.$$

The scalar localization theorem gives $g_i = 0$ for each i .

The same argument applies to a continuous tensor field $S : B \rightarrow \mathcal{V}^2$:

$$\int_{\Omega} S \, dV = 0 \text{ for every } \Omega \quad \implies \quad S = 0 \text{ in } B.$$



Example (From an Integral Balance to a Local Balance)

Suppose that a flux $\mathbf{q}$ and a source density s satisfy

$$\int_{\Omega} (s - \nabla \cdot \mathbf{q}) \, dV = 0 \quad \text{for every open } \Omega \subseteq B.$$

If $s - \nabla \cdot \mathbf{q}$ is continuous, localization gives

$$s = \nabla \cdot \mathbf{q} \quad \text{at every point of } B.$$

For heat transport, possible units are

$$[\mathbf{q}] = \frac{\text{J}}{\text{m}^2\text{s}}, \quad [s] = [\nabla \cdot \mathbf{q}] = \frac{\text{J}}{\text{m}^3\text{s}}.$$

After integration over volume, every term has units J/s.



Differential Operators in Practice

Coordinates, Dimensions, and Units



Throughout the examples, $\mathcal{F} = \{\mathbf{e}_i\}$ denotes the chosen orthonormal frame, and $x_i = [\mathbf{x}]_i$ is the i th coordinate of the point $\mathbf{x}$ in this frame. If x_i has units of length, then differentiation by x_i contributes one factor of inverse length:

$$\left[\frac{\partial f}{\partial x_i} \right] = \frac{[f]}{L}.$$

Quantity	Units
∇f	$[f]/L$
$\nabla \mathbf{v}, \nabla \cdot \mathbf{v}, \nabla \times \mathbf{v}$	$[\mathbf{v}]/L$
Δf	$[f]/L^2$
$\int_B f \, dV$	$[f]L^3$
$\int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA$	$[\mathbf{v}]L^2$

Only quantities with the same dimensions can be added or equated.



Example (Gradient of a Temperature Field)

Let

$$T(\mathbf{x}) = T_* + Gx_1 + Cx_2^2,$$

where

$$[T_*] = \text{K}, \quad [G] = \frac{\text{K}}{\text{m}}, \quad [C] = \frac{\text{K}}{\text{m}^2}.$$

Then

$$\begin{aligned} \nabla T &= \frac{\partial T}{\partial x_1} \mathbf{e}_1 + \frac{\partial T}{\partial x_2} \mathbf{e}_2 + \frac{\partial T}{\partial x_3} \mathbf{e}_3 \\ &= G\mathbf{e}_1 + 2Cx_2\mathbf{e}_2. \end{aligned}$$

Every component has units K/m.

The vector ∇T points in the direction of greatest local increase of temperature.



Example (Gradient of a Velocity Field)

Let

$$\mathbf{v}(\mathbf{x}) = ax_1\mathbf{e}_1 - ax_2\mathbf{e}_2 + bx_3\mathbf{e}_3, \quad [a] = [b] = s^{-1}.$$

Since $v_i = [\mathbf{v}]_i$,

$$[\nabla\mathbf{v}]_{ij} = v_{i,j}.$$

Thus

$$\nabla\mathbf{v} = \begin{bmatrix} a & 0 & 0 \\ 0 & -a & 0 \\ 0 & 0 & b \end{bmatrix}, \quad \nabla \cdot \mathbf{v} = \text{tr}(\nabla\mathbf{v}) = b.$$

The x_1 -stretch and x_2 -compression cancel in the trace. The x_3 -stretch produces the net local change of volume.

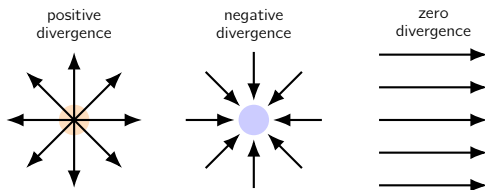
Physical Meaning of Divergence



Let $B_r(\mathbf{x})$ be the ball of radius r centered at $\mathbf{x}$. The divergence theorem gives

$$\frac{1}{\text{Vol}(B_r)} \int_{\partial B_r} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA = \frac{1}{\text{Vol}(B_r)} \int_{B_r} \nabla \cdot \mathbf{v} \, dV \xrightarrow{r \searrow 0} \nabla \cdot \mathbf{v}(\mathbf{x}).$$

The limit uses continuity of $\nabla \cdot \mathbf{v}$. Thus divergence measures outward volume flux per unit volume near $\mathbf{x}$.



For velocity, $[\nabla \cdot \mathbf{v}] = \text{s}^{-1}$.



Example (Expansion and Rotation)

Consider the two velocity fields

$$\mathbf{v}(\mathbf{x}) = \gamma(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3), \quad \mathbf{w}(\mathbf{x}) = \omega(-x_2\mathbf{e}_1 + x_1\mathbf{e}_2),$$

where $[\gamma] = [\omega] = s^{-1}$.

For the radial field,

$$\nabla \cdot \mathbf{v} = \gamma + \gamma + \gamma = 3\gamma > 0 \quad (\gamma > 0).$$

For the rotational field,

$$\nabla \cdot \mathbf{w} = \frac{\partial(-\omega x_2)}{\partial x_1} + \frac{\partial(\omega x_1)}{\partial x_2} = 0.$$

Motion can be substantial even when the divergence is zero.

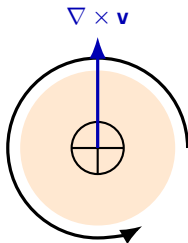


Physical Meaning of Curl

Let $A_r(\mathbf{x})$ be an oriented planar surface that shrinks to $\mathbf{x}$, with unit normal $\hat{\mathbf{n}}$. Stokes' theorem gives

$$\frac{1}{\text{Area}(A_r)} \int_{\partial A_r} \mathbf{v} \cdot \hat{\mathbf{t}} \, ds = \frac{1}{\text{Area}(A_r)} \int_{A_r} (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} \, dA \xrightarrow{r \searrow 0} (\nabla \times \mathbf{v})(\mathbf{x}) \cdot \hat{\mathbf{n}}.$$

The limit uses continuity of $\nabla \times \mathbf{v}$. Thus curl measures circulation per unit area.



a small paddle wheel rotates

For velocity, $[\nabla \times \mathbf{v}] = \text{s}^{-1}$. Its direction gives the local axis of rotation by the right-hand rule.



Example (Rigid Rotation)

Let

$$\mathbf{v}(\mathbf{x}) = \omega(-x_2\mathbf{e}_1 + x_1\mathbf{e}_2), \quad [\omega] = \text{s}^{-1}.$$

Then

$$\begin{aligned}\nabla \times \mathbf{v} &= \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \mathbf{e}_3 \\ &= (\omega - (-\omega))\mathbf{e}_3 \\ &= 2\omega\mathbf{e}_3.\end{aligned}$$

Thus the angular-velocity vector of a rigid rotation is

$$\frac{1}{2}\nabla \times \mathbf{v} = \omega\mathbf{e}_3.$$

The factor of two is important.



Example (Laplacian of a Temperature Field)

Let

$$T(\mathbf{x}) = T_* + C(x_1^2 + x_2^2 + x_3^2), \quad [T_*] = \text{K}, \quad [C] = \frac{\text{K}}{\text{m}^2}.$$

Then

$$\begin{aligned} \Delta T &= \frac{\partial^2 T}{\partial x_1^2} + \frac{\partial^2 T}{\partial x_2^2} + \frac{\partial^2 T}{\partial x_3^2} \\ &= 2C + 2C + 2C = 6C. \end{aligned}$$

Therefore $[\Delta T] = \text{K}/\text{m}^2$.

In the heat equation $T_{,t} = \kappa \Delta T$, the diffusivity has units

$$[\kappa] = \frac{\text{m}^2}{\text{s}},$$

so both sides have units K/s .

Product Rules Collected



For differentiable scalar fields ϕ, ψ and a differentiable vector field $\mathbf{v}$,

$$\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi,$$

$$\nabla \cdot (\phi\mathbf{v}) = \nabla\phi \cdot \mathbf{v} + \phi\nabla \cdot \mathbf{v},$$

$$\nabla \times (\phi\mathbf{v}) = \nabla\phi \times \mathbf{v} + \phi\nabla \times \mathbf{v},$$

$$\Delta(\phi\psi) = \phi\Delta\psi + 2\nabla\phi \cdot \nabla\psi + \psi\Delta\phi.$$

Each term in a product rule must have the same tensorial type and the same physical units.



Example (Checking the Divergence Product Rule)

Let

$$\phi(\mathbf{x}) = \frac{x_1}{L_0}, \quad \mathbf{v}(\mathbf{x}) = \frac{U}{L_0}(x_2\mathbf{e}_1 + x_1\mathbf{e}_2).$$

Here $[U] = \text{m/s}$ and $[L_0] = \text{m}$, so that $[\phi] = 1$ and $[\mathbf{v}] = \text{m/s}$.

Direct calculation gives

$$\nabla \cdot (\phi \mathbf{v}) = \frac{\partial}{\partial x_1} \left(\frac{U x_1 x_2}{L_0^2} \right) + \frac{\partial}{\partial x_2} \left(\frac{U x_1^2}{L_0^2} \right) = \frac{U x_2}{L_0^2}.$$

The product rule gives the same result:

$$\begin{aligned} \nabla \phi \cdot \mathbf{v} + \phi \nabla \cdot \mathbf{v} &= \frac{\mathbf{e}_1}{L_0} \cdot \frac{U}{L_0}(x_2\mathbf{e}_1 + x_1\mathbf{e}_2) + 0 \\ &= \frac{U x_2}{L_0^2}. \end{aligned}$$

Both sides have units s^{-1} .



Example (Checking the Curl Product Rule)

Let

$$\phi(\mathbf{x}) = \frac{x_1}{L_0}, \quad \mathbf{v}(\mathbf{x}) = \frac{Ux_2}{L_0} \mathbf{e}_3,$$

where $[U] = \text{m/s}$ and $[L_0] = \text{m}$. Direct calculation gives

$$\begin{aligned} \nabla \times (\phi \mathbf{v}) &= \nabla \times \left(\frac{Ux_1x_2}{L_0^2} \mathbf{e}_3 \right) \\ &= \frac{Ux_1}{L_0^2} \mathbf{e}_1 - \frac{Ux_2}{L_0^2} \mathbf{e}_2. \end{aligned}$$

The product rule gives the same result:

$$\begin{aligned} \nabla \phi \times \mathbf{v} + \phi \nabla \times \mathbf{v} &= \frac{\mathbf{e}_1}{L_0} \times \frac{Ux_2}{L_0} \mathbf{e}_3 + \frac{x_1}{L_0} \left(\frac{U}{L_0} \mathbf{e}_1 \right) \\ &= -\frac{Ux_2}{L_0^2} \mathbf{e}_2 + \frac{Ux_1}{L_0^2} \mathbf{e}_1. \end{aligned}$$

Here we used $\mathbf{e}_1 \times \mathbf{e}_3 = -\mathbf{e}_2$. Both sides have units s^{-1} .



Two Divergence-Theorem Calculations



Example (A Two-Dimensional Flux Calculation)

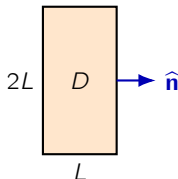
Let

$$D = \{\mathbf{x} \in \mathbb{E}^2 : 0 \leq x_1 \leq L, 0 \leq x_2 \leq 2L\}, \quad \mathbf{v}(\mathbf{x}) = \alpha(x_1^2 \mathbf{e}_1 + x_1 x_2 \mathbf{e}_2),$$

where $[L] = \text{m}$ and $[\alpha] = \text{m}^{-1}\text{s}^{-1}$, so that $[\mathbf{v}] = \text{m/s}$.

The two-dimensional divergence theorem is

$$\int_{\partial D} \mathbf{v} \cdot \hat{\mathbf{n}} \, ds = \int_D \nabla \cdot \mathbf{v} \, dA.$$





Example (A Two-Dimensional Flux Calculation (Cont.))

First compute the divergence:

$$\nabla \cdot \mathbf{v} = \frac{\partial(\alpha x_1^2)}{\partial x_1} + \frac{\partial(\alpha x_1 x_2)}{\partial x_2} = 2\alpha x_1 + \alpha x_1 = 3\alpha x_1.$$

Therefore

$$\begin{aligned} \int_D \nabla \cdot \mathbf{v} \, dA &= \int_0^L \int_0^{2L} 3\alpha x_1 \, dx_2 \, dx_1 \\ &= \int_0^L 6\alpha L x_1 \, dx_1 \\ &= 3\alpha L^3. \end{aligned}$$

The result has units m^2/s : area per unit time.



Example (A Two-Dimensional Flux Calculation (Cont.))

Compute the outward flux on each side of the rectangle:

$$x_2 = 0 : \int_0^L \mathbf{v} \cdot (-\mathbf{e}_2) \, dx_1 = 0,$$

$$x_1 = L : \int_0^{2L} \mathbf{v} \cdot \mathbf{e}_1 \, dx_2 = \int_0^{2L} \alpha L^2 \, dx_2 = 2\alpha L^3,$$

$$x_2 = 2L : \int_0^L \mathbf{v} \cdot \mathbf{e}_2 \, dx_1 = \int_0^L 2\alpha L x_1 \, dx_1 = \alpha L^3,$$

$$x_1 = 0 : \int_0^{2L} \mathbf{v} \cdot (-\mathbf{e}_1) \, dx_2 = 0.$$

The normal changes from side to side. Using one normal for the entire boundary would give the wrong answer.



Example (A Two-Dimensional Flux Calculation (Cont.))

Adding the four boundary contributions gives

$$\int_{\partial D} \mathbf{v} \cdot \hat{\mathbf{n}} \, ds = 0 + 2\alpha L^3 + \alpha L^3 + 0 = 3\alpha L^3.$$

The interior calculation gave

$$\int_D \nabla \cdot \mathbf{v} \, dA = 3\alpha L^3.$$

Hence the boundary flux equals the integrated interior divergence.

The units also agree:

$$\left[\int_{\partial D} \mathbf{v} \cdot \hat{\mathbf{n}} \, ds \right] = \frac{\text{m}^2}{\text{s}} = \left[\int_D \nabla \cdot \mathbf{v} \, dA \right].$$



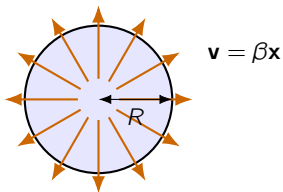
Example (A Three-Dimensional Flux Calculation)

Let

$$B = \{\mathbf{x} \in \mathbb{E}^3 : \|\mathbf{x}\| < R\}, \quad \mathbf{v}(\mathbf{x}) = \beta \mathbf{x}, \quad [R] = \text{m}, \quad [\beta] = \text{s}^{-1}.$$

The divergence theorem states

$$\int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA = \int_B \nabla \cdot \mathbf{v} \, dV.$$





Example (A Three-Dimensional Flux Calculation (Cont.))

Write

$$\mathbf{v}(\mathbf{x}) = \beta x_1 \mathbf{e}_1 + \beta x_2 \mathbf{e}_2 + \beta x_3 \mathbf{e}_3.$$

Then

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \frac{\partial(\beta x_1)}{\partial x_1} + \frac{\partial(\beta x_2)}{\partial x_2} + \frac{\partial(\beta x_3)}{\partial x_3} \\ &= \beta + \beta + \beta = 3\beta.\end{aligned}$$

Therefore

$$\begin{aligned}\int_B \nabla \cdot \mathbf{v} \, dV &= 3\beta \operatorname{Vol}(B) \\ &= 3\beta \left(\frac{4}{3} \pi R^3 \right) \\ &= 4\pi\beta R^3.\end{aligned}$$



Example (A Three-Dimensional Flux Calculation (Cont.))

On the sphere ∂B ,

$$\hat{\mathbf{n}} = \frac{\mathbf{x}}{R}, \quad \mathbf{v} = \beta \mathbf{x} = \beta R \hat{\mathbf{n}}.$$

Thus $\mathbf{v} \cdot \hat{\mathbf{n}} = \beta R$, and

$$\begin{aligned} \int_{\partial B} \mathbf{v} \cdot \hat{\mathbf{n}} \, dA &= \beta R \, \text{Area}(\partial B) \\ &= \beta R (4\pi R^2) \\ &= 4\pi \beta R^3. \end{aligned}$$

Both sides have units m^3/s . The outward flux equals the total rate of volume production inside the ball.



Two Stokes-Theorem Calculations



Example (Stokes' Theorem with Two Spanning Surfaces)

Let $[R] = m$, and let

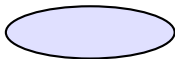
$$C = \{\mathbf{x} \in \mathbb{E}^3 : x_1^2 + x_2^2 = R^2, x_3 = 0\}$$

be oriented counterclockwise when viewed from above. Consider

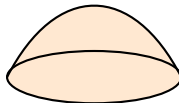
$$\mathbf{v}(\mathbf{x}) = \frac{\kappa}{2}(-x_2\mathbf{e}_1 + x_1\mathbf{e}_2), \quad [\kappa] = s^{-1}.$$

A direct calculation gives

$$\nabla \times \mathbf{v} = \kappa \mathbf{e}_3.$$



flat disk D



paraboloid-like sheet S



Example (Stokes' Theorem with Two Spanning Surfaces (Cont.))

Parameterize C by

$$\mathbf{r}(\theta) = R \cos \theta \mathbf{e}_1 + R \sin \theta \mathbf{e}_2, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$\mathbf{r}'(\theta) = R(-\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2)$$

and

$$\mathbf{v}(\mathbf{r}(\theta)) = \frac{\kappa R}{2}(-\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2).$$

Therefore

$$\begin{aligned} \int_C \mathbf{v} \cdot \hat{\mathbf{t}} \, ds &= \int_0^{2\pi} \mathbf{v}(\mathbf{r}(\theta)) \cdot \mathbf{r}'(\theta) \, d\theta \\ &= \int_0^{2\pi} \frac{\kappa R^2}{2} \, d\theta \\ &= \pi \kappa R^2. \end{aligned}$$



Example (Stokes' Theorem with Two Spanning Surfaces (Cont.))

Take

$$D = \{\mathbf{x} \in \mathbb{E}^3 : x_1^2 + x_2^2 \leq R^2, x_3 = 0\}, \quad \hat{\mathbf{n}} = \mathbf{e}_3.$$

This is a planar region in $\mathbb{R}^2$, viewed as a surface in $\mathbb{E}^3$. Then

$$\begin{aligned} \int_D (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} \, dA &= \int_D (\kappa \mathbf{e}_3) \cdot \mathbf{e}_3 \, dA \\ &= \kappa \text{Area}(D) \\ &= \pi \kappa R^2. \end{aligned}$$

This agrees with the circulation around C . Both sides have units m^2/s .



Example (Stokes' Theorem with Two Spanning Surfaces (Cont.))

Now take

$$S: \quad x_3 = h \left(1 - \frac{x_1^2 + x_2^2}{R^2} \right), \quad x_1^2 + x_2^2 \leq R^2.$$

Here $[h] = \text{m}$, so the two sides of the surface equation have the same units. Its boundary is still the circle C . An upward-oriented parameterization is

$$\mathbf{x}(r, \theta) = r \cos \theta \mathbf{e}_1 + r \sin \theta \mathbf{e}_2 + h \left(1 - \frac{r^2}{R^2} \right) \mathbf{e}_3,$$

where

$$0 \leq r \leq R, \quad 0 \leq \theta \leq 2\pi.$$

At $r = R$, the height is zero, so the parameterized boundary is exactly C .



Example (Stokes' Theorem with Two Spanning Surfaces (Cont.))

Differentiate the parameterization:

$$\mathbf{x}_{,r} = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2 - \frac{2hr}{R^2} \mathbf{e}_3,$$

$$\mathbf{x}_{,\theta} = -r \sin \theta \mathbf{e}_1 + r \cos \theta \mathbf{e}_2.$$

Hence

$$\mathbf{x}_{,r} \times \mathbf{x}_{,\theta} = \frac{2hr^2}{R^2} \cos \theta \mathbf{e}_1 + \frac{2hr^2}{R^2} \sin \theta \mathbf{e}_2 + r \mathbf{e}_3.$$

The positive $\mathbf{e}_3$ component confirms the upward orientation. Also,

$$\hat{\mathbf{n}} \, dA = (\mathbf{x}_{,r} \times \mathbf{x}_{,\theta}) \, dr \, d\theta.$$



Example (Stokes' Theorem with Two Spanning Surfaces (Cont.))

Since $\nabla \times \mathbf{v} = \kappa \mathbf{e}_3$,

$$\begin{aligned} \int_S (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} \, dA &= \int_0^{2\pi} \int_0^R \kappa \mathbf{e}_3 \cdot (\mathbf{x}_{,r} \times \mathbf{x}_{,\theta}) \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^R \kappa r \, dr \, d\theta \\ &= \pi \kappa R^2. \end{aligned}$$

The line integral, the flat-disk integral, and the curved-sheet integral all agree. The height h changes the surface, but it does not change the answer:

$$\int_C \mathbf{v} \cdot \hat{\mathbf{t}} \, ds = \int_D (\nabla \times \mathbf{v}) \cdot \mathbf{e}_3 \, dA = \int_S (\nabla \times \mathbf{v}) \cdot \hat{\mathbf{n}} \, dA.$$